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Question 1: How do formal methods benefit safety-critical systems?

- A. By eliminating the need for human oversight
- B. By reducing the number of errors
- C. By increasing system complexity
- D. By delaying the development process

Answer: B

Question 2: Why is precision important in software specification?

- A. To simplify the user interface
- B. To increase development speed
- C. To avoid serious errors later
- D. To reduce the cost of software

Answer: C

Question 3: What is the primary purpose of using UML in software development?

- A. Conducting user training
- B. Performing system maintenance
- C. Writing code
- D. Design and requirements analysis

Answer: D

Question 4: How does model checking verify system properties?

- A. Through an exhaustive search of all possible states
- B. By comparing code with user manuals
- C. By executing random test cases
- D. By using AI algorithms to predict errors

Answer: A

Question 5: Which option is a key characteristic of high-quality software?

- A. Complexity
- B. Correctness
- C. High cost
- D. Limited functionality

Answer: B

Question 6: Which phase follows the implementation phase in software development?

- A. Verification
- B. Requirements analysis
- C. Design
- D. Maintenance

Answer: D

Question 7: How can formal methods improve software maintainability?

- A. By reducing initial development costs
- B. By providing a precise and consistent specification
- C. By simplifying the user interface
- D. By limiting software features

Answer: B

Question 8: What is one advantage of human-directed proof in formal methods?

- A. Eliminates all errors

- B. Faster than automated proof
- C. Less mathematical sophistication required
- D. Improved understanding of the system

Answer: D

Question 9: What type of semantics uses preconditions and postconditions?

- A. Operational semantics
- B. Denotational semantics
- C. Functional semantics
- D. Axiomatic semantics

Answer: D

Question 10: Which formalism is used for the specification of reactive systems?

- A. Statecharts
- B. Algebraic specification
- C. Predicate logic
- D. UML

Answer: A

Question 11: Which statement is true about system testing?

- A. It focuses on individual modules
- B. It is only concerned with user interfaces
- C. It is done by software developers
- D. It tests a fully integrated software product

Answer: D

Question 12: What is the focus of validation during software testing?

- A. Increasing software complexity
- B. Building the product correctly
- C. Building the right solution
- D. Reducing software size

Answer: C

Question 13: Which proofing method swaps the result into the condition and negates both?

- A. Induction Proof
- B. Contra-positive Proof
- C. Direct Proof
- D. Sequential Proof

Answer: B

Question 14: What does a test plan typically contain?

- A. A description of hardware specifications
- B. A sequence number for each test item
- C. A list of user requirements
- D. A summary of software features

Answer: B

Question 15: How is the correctness of a software module tested in unit testing?

- A. By integrating with external systems
- B. By testing all modules together
- C. By focusing on an individual module
- D. By testing the user interface

Answer: C

Question 16: A conjecture is best described as:

- A. A statement taken for granted as true
- B. A proven statement used to prove other statements

- C. A statement that contradicts itself
- D. A statement believed to be true but not yet proven

Answer: D

Question 17: What does a mathematical proof aim to establish?

- A. The popularity of a statement
- B. The correctness or truth of a statement
- C. The complexity of a statement
- D. The length of a statement

Answer: B

Question 18: Which testing approach starts from a general level down to specific details?

- A. Top Down
- B. Outside In
- C. Inside Out
- D. Bottom Up

Answer: A

Question 19: Which type of testing is done by the end user in a controlled environment using real-life data?

- A. Unit Testing
- B. System Testing
- C. User Acceptance Testing
- D. Integration Testing

Answer: C

Question 20: What is the main goal of stress testing in software development?

- A. To break the system and determine safe levels
- B. To validate software requirements
- C. To test user interface design
- D. To verify algorithm efficiency

Answer: A

Question 21: What does procedural cohesion focus on in software design?

- A. Grouping elements executed sequentially to perform a task
- B. Grouping elements based on time
- C. Grouping unrelated elements
- D. Grouping elements based on data usage

Answer: A

Question 22: What is an example of concurrency in software design?

- A. Saving data to a database
- B. Writing code for a single module
- C. Debugging a program
- D. Spellcheck running alongside a word processor

Answer: D

Question 23: How does concurrency benefit software execution?

- A. It allows parallel execution of multiple modules
- B. It increases dependency between modules
- C. It combines all modules into one process
- D. It ensures sequential execution of code

Answer: A

Question 24: Which component of a DFD represents the source and destination of information?

- A. Data Storage
- B. Process

- C. Entities
- D. Data Flow

Answer: C

Question 25: How does a structure chart differ from a DFD?

- A. A structure chart is less detailed than a DFD
- B. Both are identical in representation
- C. A DFD shows hierarchical structure while a structure chart does not
- D. A structure chart represents the system in more detail than a DFD

Answer: D

Question 26: Why is design verification important in software design?

- A. To remove all modules
- B. To ensure accuracy and quality before implementation
- C. To finalize the user interface
- D. To begin coding without specifications

Answer: B

Question 27: Explain the concept of modularization in software design.

- A. Combining all modules into a single unit
- B. Dividing a software system into multiple discrete and independent modules
- C. Removing unnecessary modules from the system
- D. Testing each module separately

Answer: B

Question 28: Why is it beneficial to have components with high cohesion?

- A. They increase module dependency
- B. They can be reused in different parts of the software
- C. They complicate the software design process
- D. They are difficult to maintain

Answer: B

Question 29: Which document is created to assess user requirements before coding?

- A. Implementation Plan
- B. Software Requirement Specification (SRS)
- C. User Manual
- D. Design Verification Document

Answer: B

Question 30: Describe the process of creating a Level 0 Data Flow Diagram (DFD).

- A. It includes control and branch elements
- B. It depicts the entire information system as one diagram without underlying details
- C. It focuses on the physical implementation of data flow
- D. It shows detailed flow of data within modules

Answer: B